

Raphael Catenacci

Montréal – Canada

☎ +1 (263) 378-3774 • ✉ raphael.catenacci@icloud.com • 🌐 raphaelcatenacci.com

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Education

Université de Montréal <i>M.Sc. Exchange in Mathematical and Computational Finance</i> ○ One-year academic exchange as part of the ENSAE Paris engineering curriculum.	Montréal, Canada 2026 – 2027
ENSAE Paris <i>Engineering School of the Institut Polytechnique de Paris</i> ○ Relevant Courses: Financials Mathematics, Stochastic Processes, Machine Learning Fundamentals, Financial Instruments, Time Series, Statistical Simulations and Monte Carlo Methods, Python for Data Science (Scikit-Learn, Geopandas, API and Webscraping).	Palaiseau, France 2023 – 2027
International Center of Valbonne <i>Mathematics and Physics Preparatory Classes</i>	Sophia Antipolis, France 2021 – 2023

Professional Experience

Jukoï Capital <i>Quantitative Researcher – Part-time</i> ○ Developing an end-to-end algorithmic trading platform, from financial data ingestion and storage to strategy research, backtesting, and performance monitoring. ○ Designing a Bronze, Silver, and Gold data lakehouse to standardize market, macroeconomic, and alternative data while maintaining metadata, data quality, and lineage. Integrating specialized AI agents to extract and structure information from unstructured sources, including financial reports, audio, video and more. ○ Building reusable quantitative research pipelines for feature engineering, signal generation, portfolio construction, risk analysis, and strategy evaluation. ○ Technology stack: Python, Parquet, Apache Iceberg, DuckDB, Polars, Dagster, SeaweedFS/S3, Docker.	Monaco / Remote Sep 2026 – Present
Jukoï Capital <i>Quant Researcher off-cycle</i> ○ Developed a fully automated, provider-agnostic financial data infrastructure for scalable market data ingestion and processing. (Stack: Bloomberg, Django, PostgreSQL, FastAPI, Uvicorn) ○ Automated the financial instrument lifecycle management process, including corporate actions handling, pricing updates, event processing and portfolio synchronization. (Stack: Quantlib) ○ Implemented risk calculations, stress testing, exposure measurement, and pricing models for trading portfolios, with seamless integration into a web-based financial application.	Monaco Jun 2025 – July 2026
AXA France <i>Data Scientist Intern</i> ○ Developed algorithms within the financial compliance team. ○ Built an artificial neural network using the Transformers library. Used Pandas, Scikit-Learn, and Pytorch for Machine Learning.	Paris Jun – Sep 2024

Skills

Programming and Software: Python (Advanced), JavaScript, HTML/CSS, SQL, R,git, Docker, Microsoft Office, LaTeX
Languages: French (native), English (Fluent) – MaltaLingua School of English certificate

Projects

2025: Applied Statistics project for ENGIE: Developed a CatBOOST Machine Learning model to forecast French national gas consumption. Performed fine-tuning of hyperparameters and leveraged AutoML tools to optimize model performance.

2025: Financial Machine Learning Automatic Trading Algorithm: Designed and implemented various trading strategies using machine learning techniques. Conducted performance evaluation with the calculation of key financial metrics such as Sharpe Ratios, drawdowns, and other risk-adjusted performance indicators.

2024: Real Estate Price forecast: Forecasting real estate prices using Machine Learning algorithms and public datasets.

Extra-Curricular Activities

Sports: BMX Racing (2019 UCI World Ranking U18: 9th)
Mathematics: Participation in the 2021 French general mathematics competition